The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

Abstract and storyline

Abstract

Temporal-interference (TI) stimulation requires field-to-membrane access, nonlinear generation of a difference-frequency component absent from the field, and amplification of that component. We connect these stages for weak perturbations of ongoing activity. Single-neuron TI models already demonstrate axonal ion-channel rectification. We represent candidate fast access through axonal compartments, including nodes and the axon initial segment, by a short effective membrane time. Periaxonal circuitry motivates separate axonal and somatic voltage responses. We analyze mixing and recurrent gain in a Jansen–Rit column, an exact quadratic integrate-and-fire mean field, and its spiking network. For a smooth response with a nonzero second-order term, cellular contributions combine into a beat proportional to the product of the effective carrier amplitudes. A static rate curve mixes through its curvature; the exact mean field recovers this description when membrane relaxation is fast relative to the input, leaving dynamic synapses. Otherwise mixing depends on frequency. Membrane attenuation enters once, through access. Near a Hopf bifurcation, the network amplifies the generated beat as $1/\gamma$ within linear response, where γ is modal damping. Coupling sweeps separate this gain from leading-order beat generation. Rate modulation grows faster than the mean-rate change near resonance. Spiking simulations distinguish beat modulation below the oscillatory threshold from gamma induced when rectification crosses it. An equal-power mean-field carrier control separates induction from the beat response. Fast access, cellular mixing, and slow circuit resonance form a consistent candidate mechanism; its field-to-voltage transfer and response at human field strengths require calibration. The product law makes envelope-only dosimetry conditional on transduction. Within linear response, TI and direct low-frequency stimulation share the amplifier when coupling, readout, and state are matched, so resonance alone gives TI no efficiency advantage.

The article, technical note, and storyline share the four abstract macros in `planning/tn0484_abstract.tex`. The JNE article is the working edition; only its run-in headings differ. Draft10 implements the placement decisions relayed on 5 October: LaNMM methods join their results in Appendix E; transduction laws remain in Theory, with their measurement procedure in Methods; the equal-RMS carrier control moves from S13 to Figure 11. All scientific content is retained. Draft11 corrects notation, citation scope, and the access argument; the separate computational Landau findings await integration. The closing record gives the reasons for the placements.

Problem and theory

First argument block: Sections 1–2

A two-tone field carries no power at its beat frequency. Access converts the applied field into local membrane polarization; nonlinear detection generates the beat; the network selects and amplifies that generated input. The Introduction develops these stages in that order, with their mechanisms and evidence. Axonal TI rectification is established in prior models; Short effective membrane times were already tested in a TI gamma-network model (Esmaeilpour et al., 2021). Our contribution is the separation of mixing from recurrent gain, the exact QIF mean field, the product law, and the shared TI/tACS amplifier. Fast axonal compartments are the access hypothesis: Cohen's periaxonal circuit supports distinct nodal and internodal voltage dynamics; the AIS is a separate candidate with a high sodium-channel density. Neither observation fixes a TI charging time. Luff supplies evidence for cellular mixing, while cellular resonance and recurrent amplification remain distinct. QIF connects the effective weak-input account to a neuron model; its static transfer is an adiabatic limit, not a consequence of small amplitude alone.

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§1 Introduction	The two-tone field (Eq. 1) has lines only at the applied tones; a linear system cannot create the beat. Cellular mixing exists (MRG axon model with Hodgkin–Huxley dynamics; Luff's subthreshold mixer at 5 kHz, TTX-sensitive). Primate timing changes establish modulation; synaptic blockade and network simulations support circuit involvement. Retain weak modulation of ongoing activity as the scope, with direct stimulation thresholds and block outside it. Weak input can perturb spiking neurons; linear gain requires stability under the rectified mean shift. Introduce fast axonal access before detection: AIS and nodes are candidates; Cohen's submyelin circuit explains distinct axonal voltage modes, not a measured TI transfer. Cellular contributions enter an effective quadratic response, and cellular tuning remains distinct from the recurrent amplifier. Retain the candidate positively and name its calibration gap.	States the physical sequence: field-to-membrane access, beat generation, then amplification. Places each reference beside the stage it supports. Announces the ladder (JR column, QIF mean field, spiking network, laminar column) once. The transfer curve must be defined before its curvature can mix, so §2.1 follows.	Fig. 1 (receiver concept). Cohen 2020; Kole 2008 (AIS); Mirzakhalili 2020; Luff 2024 (mixer); Vieira 2024; Caldas-Martinez 2024; CFC and HAM companion papers.

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§2.1 Population transfer and the slow-synapse limit	JR synaptic filter and sigmoid (Eqs. 2–3); NMM1 with the direct field term $\Lambda[E]$ (Eq. 3). The MPR mean field (Eq. 4) and its static curve $F(Q)$ (Eq. 5), inflection at $Q = \Delta/\sqrt{3}$, nonlinear even at zero recurrence and unbounded. Retain its derivation from the single-cell square-root rate and threshold recruitment. The slow-synapse reduction (Eq. 6) and its two conditions: $\tau_{\mathrm{rel}} \ll \tau_s$ makes the transfer static for synaptic input; $\omega_c \tau_{\mathrm{rel}} \ll 1$ is the stronger condition for a carrier. Effective curvature collects static mixing; the coefficient's value and dynamics remain model-specific. Cellular and circuit resonance are distinct; the reduction removes membrane-dependent modes.	Defines what a transfer curve is and when it is static, so that curvature can be a mixer. Introduces the time-scale distinction that the Discussion and Conclusion reuse. The field must reach the curve, which §2.2 supplies.	Rosetta Stone (Ruffini 2025); Montbrió 2015; Clusella 2023; de Palma Aristides 2026 (WP0242, unpublished); Hutcheon and Yarom 2000.
§2.2 From applied field to a squared-envelope beat	Quasistatic field; polarization $s(t)$ through parallel entry points with access filters (Eqs. 7–8), one compartment response counted once. Taylor expansion (Eqs. 9–10) with the truncation condition $\lvert s\rvert \ll 1/\rho \approx 1.8$ mV. The square of the envelope (Eq. 11): A^2 is sinusoidal, A is a rectified cosine. The AM surrogate (Eq. 12), where it is used and where it is not. The effective dynamic coefficient $\chi^{(2)}$ combines cellular contributions and reduces to $\Sigma''/2$ in the static case. Retain smoothness and nonzero second-order response, its possible frequency dependence, and the DC term. Static predictions: ε^2 scaling, Σ'' law, carrier independence at fixed polarization.	Derives the beat and its amplitude: the first result. Fixes the waveform discipline (two tones for every waveform-sensitive claim). The derivation assumes a static curve, so §2.3 asks what a dynamic population does.	Figs. 3 and 6 (two-tone runs); Fig. 6d (surrogate check). HAM paper (Volterra kernel); Galan and Gadea 2023.
§2.3 The exact mean field connects static and dynamic mixing	Single-carrier DC coefficient $C(Q,\omega)$ (Eq. 13), with limits $F''/4$ and $F'/(2\omega^2)$ (Eq. 14), sign change at ω_z above the inflection. Fixed-excursion coefficient $K(\omega)$ (Eq. 15): the $1/\omega^2$ is the membrane, not a detector filter. The PING v_E^2 source matches the propagated second-order beat response to 1.1%, 0.1%, 0.03% at 300, 1000, 2000 Hz at four stable points. This is not a full finite-amplitude sweep validation. Carrier and beat take different routes to the rate (Table 1).	Connects the static and dynamic QIF descriptions, with membrane access counted once. Retain the distinction between the DC calculation and the two-tone beat source. Whatever the detector, its envelope law has observable signatures, which §2.4 derives.	Fig. 7; Table 1. Code: `static_vs_dynamic_check.py`, `mpr_twotone_lines_check.py`.

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§2.4 Transduction class and its signatures	Cycle average (Eq. 16) is even in A; nonzero curvature gives a leading square law for a smooth memoryless transfer. Specified ideal rectifiers follow A; a step detector follows neither (harmonic ratio 0.151). Harmonic ladder 1/5, 3/35, 1/21 and r_{det} (Eq. 17), whose inverse needs the relative sign or a branch choice. Smooth square-law detection has a vanishing weak-field harmonic ratio; a threshold detector instead has an onset and an amplitude-dependent ratio. Each harmonic is multiplied by its network gain. Excursion map (Eq. 18): product law, 2× versus 40× example; not an account of focality. DC intensity sums without cross-frequency cancellation after beat averaging: 202× superficial/deep contrast in the example; local beat/DC at most one. Retain the maximum-principle assumptions (homogeneous isotropic conductor, fixed coupling), the distinction from spatially weighted physiological response, and isotropic-orientation cancellation of the beat for orthogonal fields. Equal-RMS equivalence is scoped to second order with equal DC coefficients at the carriers. Spatial response (Eq. 19) and signed vector coupling.	Derives the detector's amplitude, harmonic, and spatial signatures and separates spatial from dynamical selectivity. Generation gives a weak drive; §2.5 explains what makes it large.	App. D (bound and step detector); Fig. 6b (slope 1.99); Table 3 (predictions).
§2.5 Amplification: synaptic low-pass and network resonance	Synaptic transfer H_s (16 Hz corner; 0.7 at 10 Hz, 2.5% at 100 Hz). One-mode reduction, $\mu_{\pm} = -\gamma \pm i\omega_0$, $\mathcal{Q} = \omega_0/2\gamma$, gain $\propto 1/\gamma$ (Eq. 20); entrainment above the Hopf. Receiver equation (Eq. 21) with D and G and their normalization for JR and QIF. The cubic normal form has no quadratic mixer.	Supplies the second factor of the receiver and the claim that a large susceptibility is not evidence of detection. The carrier must still reach the mixer, which §2.6 quantifies.	Fig. 5; Figs. 4 and 9; App. C.
§2.6 Carrier access depends on the effective membrane response	Point-membrane H_m (Eq. 22); $\lvert H_m\rvert^2 \approx 10^{-4}$ at 1 kHz for 16 ms, 0.39 and 0.14 at 1 and 2 kHz for 0.2 ms, labeled as assumed. Table 2 separates somatic measurements, Cohen's approximately 1 ms minimum axonal decay, assumed fast access, prior 1/0.4 ms TI network parameters, channel kinetics, and synaptic filtering. Cohen is cited here as well as in the Introduction and Discussion. AC space constant (Eq. 23), one tenth at 1 kHz; at a long sealed cable end this gives quadratic suppression of about 10^{-2}. Channel kinetics belong in the effective mixing response, counted once; no universal gate cutoff follows from QIF. Carrier sweeps constrain access plus mixing; beat sweeps probe the amplifier. The plotted simulations retain finite membrane times; fast QIF relaxation is an analytic limit.	Separates carrier independence at fixed polarization from the applied-field roll-off and connects the fast-compartment hypothesis to a reduced-model sensitivity calculation. The detector coefficient also depends on state, which §2.7 takes up.	Figs. S5, S6, S9; Deitcher 2017; Eyal 2016; Cohen 2020; Discussion comparison with Luff and Semenov.

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§2.7 Operating-point dependence	Signed PEIX (Eq. 24) is zero at the inflection and approaches ±1 in saturation; the normalized slope is $1 - \lvert\text{PEIX}\rvert$. Static mixing vanishes at the inflection; slope and curvature play distinct roles in the memoryless detector. The fast-carrier QIF DC coefficient follows F'; a beat also needs its finite-frequency susceptibility. State shifts scale, and in JR can flip, the response. Retain the slope-versus-curvature distinction.	Turns the detector coefficient into a brain-state prediction and marks which part is model-specific. The theory is complete; Methods specify the simulations that test it.	Fig. 2b; Fig. 6a; CFC paper (PEIX).

JR, Jansen–Rit; MRG, McIntyre–Richardson–Grill; QIF, quadratic integrate-and-fire; MPR, Montbrió–Pazó–Roxin; NMM1/NMM2, sigmoid and exact mean-field neural mass models; PING, pyramidal–interneuron network gamma; AM, amplitude modulation; PEIX, Population Excitation–Inhibition Index; CFC, cross-frequency coupling; HAM, hierarchical amplitude modulation.

Simulations

Second argument block: Sections 3–4

The JR column tests the predictions of §2.2, and the exact mean field derives generation and amplification from QIF dynamics. Finite-amplitude spiking simulations illustrate the coexistence of beat-locked and gamma activity. Methods stand apart because the lock-in read-out and the pinned-operating-point protocol recur across models.

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§3.1 Amplitude scaling and harmonic measurements	Lock-in (Eq. 25): Hann window, weighted-mean subtraction, physical-amplitude normalization, and finite-window leakage limits. Amplitude-slope test at fixed coupling and linear downstream gain. Harmonic ratio corrected by $\lvert G(2\Omega)/G(\Omega)\rvert$: at resonance and $\mathcal{Q} = 10$, 20% reads 0.7%. Off resonance at $\Delta f \simeq f_0/4$, the correction is 1.22–1.25; an independent weak tACS sweep provides the alternative deconvolution.	Collects the measurement procedure after its theoretical predictions, preserving the estimator and the restrictions on interpreting a measured harmonic ratio.	Fig. 6b; §2.4; §4.4 and §4.9.
§3.2 Jansen–Rit: integration, lock-in, controls	Field enters only the pyramidal sigmoid; the read-out $v = y_1 - y_2$ is not fed the field. RK4 steps with $< 1\%$ convergence; settling and windows. Upper Hopf at $p \approx 315.70$, $f_0 \approx 11.16$ Hz; alpha cycle between Hopf points near 90 and 316. The SNIC near 114 belongs to the larger cycle. The estimator in §3.1 retains DC subtraction to 10^{-8}. Open-loop detector versus closed-loop column; square law, linearization, finite-difference checks.	Specifies the JR integrator and lock-in; run-specific reproduction limits remain recorded.	Fig. 2; Grimbert and Faugeras 2006; `run_all.py`.
§3.3–3.4 NMM2 PING and QIF network	PING (Eq. 26): weights $(1, 1, 1, 2)$, $\bar{\eta}_H = 1.110$, $f_0 = 54$ Hz, field as current $I_E = PE$ so the membrane computes access with $\tau_m = \tau_E$; two criticality knobs. QIF: $N = 8000$ per population, Lorentzian quantiles, beats 55 and 42 Hz at $\bar{\eta} = 0.85/1.5$, defined by the field-free states; $A_f = 35$ is finite amplitude, with carrier excursion about 1.2 per tone. Modulation depth and pairwise phase consistency; the 45% threshold sensitivity of phase concentration.	Fixes the models whose results follow and states which comparisons are perturbative. The LaNMM method now accompanies its results in App. E.1.	Eq. 26; Rosetta Stone PING parameters; Vinck 2010.
§3.5–3.6 Coupling sweeps, tACS control, vocabulary	Pinning: solve $p(C)$ for fixed v^*, read γ, ω_0 from the Jacobian; tACS control at $\varepsilon = 0.1$ mV. TI uses the AM surrogate at 0.3 mV and $m = 1$; pinning is field-free and does not cancel driven mean shifts. Forced response, phase capture, and amplitude gating are distinct. The maps measure frequency dominance ($\lvert f_{\text{out}} - \Delta f\rvert < 0.3$ Hz); the rate fold does not uniquely identify gamma-amplitude gating.	The pinning protocol holds the local detector curvature fixed; the vocabulary keeps “entrainment” from covering gating.	App. F (metrics); Pikovsky 2001; Canolty 2010; Tort 2010.

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§4.1–4.2 The sigmoid demodulates; the beat drives a resonance	An alpha line from two tones with no alpha power in the input. Approximately Lorentzian response growing $0.23 \rightarrow 0.43 \rightarrow 0.70$ mV toward the Hopf ($1/\gamma$); above the Hopf, frequency dominance read by output frequency, without a phase-capture test.	The existence result: the beat is synthesized and amplified.	Figs. 3, 4; Fig. S4.
§4.3–4.4 Carrier independence, kHz roll-off, curvature law	Flat response over 25–300 Hz at fixed polarization; $1/f_c^2$ roll-off for 16 ms, retained access for 0.2 ms; two tones at 2 kHz with 2.2×10^{-13} mV input spectral residue. Curvature law to 0.9%; slope 1.99; linearization collapses the response 634×; two-tone/AM ratio 1.0000 at 0.02 mV and 0.988 at 1 mV.	Verifies the static detector predictions. Retain the one-point open-loop scope of the surrogate check; recurrent AM sweeps are explicitly qualified.	Figs. 5, 6; Figs. S6, S9; Esmaeilpour 2020.
§4.5 Exact mean-field confirmation	PING generates and amplifies the beat through v_E^2; diagonal ridge with operating point; the fast-carrier source predicts the second-order response within 1.1% at four tested stable states; this does not bound source variation across every sweep.	The receiver is not an artifact of the sigmoid.	Figs. 7, 8; Fig. S8; Clusella 2023.
§4.6 Coupling sets the gain at fixed detection	Pinned J-curve: $\sim$ 30× growth, $1/\gamma$ then saturation; the same gain–γ trend via p; stable-focus side only. Frequency and modal projections vary too. The TI/tACS ratio cancels the common transfer in linear response (0.143 predicted); the fixed-amplitude cache ratio rises to 0.24 and 0.47 near the Hopf. Retain the unequal-input and field-free-pinning qualifications; matched-source tACS follows TI with ratios 0.99, 0.94, 0.92 at three tested couplings. Both driven means can shift. Do not infer a universal efficiency gain. NMM2 J-curve uses an unpinned fast-carrier normalization, qualified at finite carrier. The full forced endpoint is stable despite the DC-only screen; retain the measured curve.	The mesoscale claim: network gain grows near a weakly damped mode while the JR local curvature is fixed; parameter paths need not have identical finite-distance gain.	Fig. 9; Fig. S10; Clusella 2023.

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§4.7–4.8 Timing over rate; QIF amplitude regimes	Modulation grows 13.5× (26 to 346 mHz) against a mean shift of 5.8× (−1.3 to −7.7 mHz); ratio 19 far from the Hopf, 45–66 near it. QIF: at $\bar{\eta} = 0.85$, stimulation induces gamma through a rectified E-drive shift beyond the E-only Hopf; at 1.5, gamma dominates a smaller beat component. Phase folding suppresses that gamma. Retain sidebands, beat depths 0.29/0.23, rate +8–11%, and weak per-neuron locking; these runs do not test weak-input scaling or establish gating. At $A_f = 10$, the stable-state spiking depth is 0.093 with rate +2.8%; retain the cutoff sensitivity: depths 0.093/0.127/0.139 at spike cutoffs 100/200/400, versus mean-field 0.128. At half step, the higher-cutoff depths are 0.121/0.141; cutoff dependence persists and quantitative convergence remains unestablished. In the equal-RMS mean-field comparison, both waveforms induce gamma at the larger amplitude; TI adds a resolved beat component. A fourfold longer window suppresses the carrier's apparent 55 Hz line below 10^{-4}. The complementary comparison is Figure 11 (formerly S13); retain the original threshold-crossing and autonomous-gamma examples in Fig. 10 and S12.	Larger modulation than mean-rate change at mesoscale and microscale, qualitatively consistent with Vieira; retain the finite-amplitude scope, nonzero mean shift, and weak per-neuron locking.	Figs. 10–11; Figs. S11–S12; Vieira 2024.
§4.9 Falsifiable predictions	Four predictions with tests: access versus mixing by carrier sweep; state and carrier dependence; transduction class by amplitude slope and harmonic ratio off resonance; frequency selectivity. Table 3 against existing evidence; synaptic block tests network involvement without excluding a cellular detector.	Converts the mechanism into falsifiable statements and bridges to the Discussion.	Table 3; Vieira 2024; Caldas-Martinez 2024; Opancar 2025.

LaNMM, laminar neural mass model; tACS, transcranial alternating-current stimulation; SNIC, saddle-node on an invariant circle; RK4, fourth-order Runge–Kutta.

Interpretation and supporting material

Third block: Sections 5–6 and Appendices A–G

The Discussion changes level from model results to physiology and states each scope limit at the claim it limits. The appendices hold the notation, the model equations, the two bounds, the laminar extension, and the metrics.

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§5 Discussion	Weak-field cellular contributions enter an effective quadratic response; QIF provides a realization, without requiring channel identification. Weak amplitude and fast relaxation justify distinct approximations. Access candidates include nodes and AIS; strong terminal polarization alone does not establish fast charging (RC independent of area). Cifra excludes fields below 0.1 kV/m from the review's scope, not by a universal threshold. Cohen's periaxonal space ($\sim$ 1 ms fitted decay), Luff's 1 ms RC, Semenov's 0.7 μs charging: a 10–100$\times$ shorter time is a hypothesis. Dose map: excursion versus product and the DC term, referencing the derivation in §2.4. The equal-RMS carrier comparison separates the operating-point shift from the beat. Intensity changes the mechanism. Shared amplifier with tACS; temporal, not spatial, criticality. Couplings 0.12–0.5 mV per V/m; $\eta_{\mathrm{det}} \simeq |\Sigma''|\varepsilon/4\Sigma'$; $\mathcal{Q} \sim 10$. Vieira's unmatched-current two-pair TI result (0.34, CI 0.04–0.65) is separate from the matched-depth AM control (PLV 0.10 versus 0.20). The illustrative 0.1 mV budget gives 0.8% at $v^* = 8.30$ mV; drive and PLV are different observables. The 0.348 supremum and its scope. Limitations: Plovie 2025, noise, columns, cascaded synapses cannot solve access, mixed human results.	Reinterprets each result in physiological terms and places every caveat where it bites. Retain each level (anatomical access, mixing, time scales, dosimetry, intensity, efficiency, scale, data, bound, limits), with access preceding detection. Retain matched-peak versus equal-RMS distinctions, and matching of input coupling, readout, and state for the shared amplifier.	Fig. S3 (tACS); Cohen 2020; Luff 2024 (PWM-TI); Semenov 2025; Vieira 2024; App. D.
§6 Conclusion	Effective quadratic mixing generates, resonance amplifies, and a large response does not identify a microscopic mixer. Static and dynamic descriptions connected by time scales; each filter once. Product law; shared amplifier within linear response; fast access represented by an effective membrane time, building on prior axonal TI mechanisms. Its TI-field transfer and human-field response await calibration.	Restates access, detection, time-scale reduction, and amplification without new claims.	—
App. A–B Notation and units; model equations	Table 4 (symbols, with the reused-glyph head note); JR equations (Eq. 27) and Table 5.	Recovery aids; definitions remain local.	—

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App. C–D Cubic normal form; efficiency bound	Stuart–Landau has no second-order source (Eq. 30); numerical control at the floor. $\eta_{\mathrm{det}} \leq \rho\varepsilon/4$ (Eq. 31); family supremum 0.3479 at $\delta/2\varepsilon = 0.523$ (Eq. 33), a step detector with harmonic ratio 0.151; the microvolt width is an idealization, and the maximum is numerical, not certified.	C supports §2.5 (susceptibility is not detection); D supports §2.4 and the Discussion's efficiency paragraphs.	`threshold_bound_check.py`.
App. E.1–E.2 Laminar methods and low-carrier results	Method: field as a signed polarization inside the sigmoid (Eq. 34), with no rate clipping; a clip would itself rectify. RK4 integration, alpha-band power including autonomous activity, signed power maps, and the 17-beat lock-in mean with 3/10 s windows and field-free background controls. Results: frequency selection at 25–80 Hz carriers up to 3 mV, outside the perturbative regime; 95%/97% of P_1-driven grids below baseline. Off-alpha mean lock-in peaks at 39 Hz for superficial drive and 25 Hz for direct drive, with unequal amplitudes. Retain the 3/10 s comparison, field-free background, and limits of the near-alpha readout. The old 7%/25% claims are replaced by the reproducible carrier-profile table.	Places the extension's method beside its results. A band-limited intermediary makes the carrier choice matter; these results do not carry the kHz argument.	Fig. S1, S2, S7; Gordon 2019.
App. F–G Vocabulary metrics; supplementary figures	Dominance screen versus capture test; folded-rate metric versus gamma-amplitude gating; frequency-dominance protocol. Figures S2–S12 support §4.2–4.8 and the Discussion.	Metrics behind the three terms; the figure bank for the JNE supplement.	Figs. S2–S12.

Placement decisions, 5 October 2026. K authorized moving LaNMM methods beside their results in Appendix E and promoting the equal-RMS carrier control from S13 to Figure 11 at its use in the QIF/PING results; S3, S10, and S11 remain supplementary. The transduction split preserves the seven Theory subsections: §2.4 derives the envelope and square-law signatures and their spatial maps without a forward dependency on a measurement procedure. Methods §3.1 collects the lock-in estimator, amplitude-slope measurement, and harmonic-gain correction; Results refers to that procedure. All 23 figures, equations, examples, and qualifications remain. The reading baseline is draft9 including the competing-interests statement (`8016065`). Coauthor confirmation, remaining funding declarations, and release alignment remain pending; see `todo.md`.